

Too Open to be Secure: An Evaluation of OpenNIC DNS Services and Domains

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Abstract—OpenNIC is a community-governed top-level Network Information Center whose DNS resolvers can resolve domains from the ICANN root, OpenNIC-operated namespaces, and its peered namespaces. Unfortunately, OpenNIC domains have been abused for malicious activities like botnets, yet little research has systematically examined this ecosystem. This paper presents the first in-depth study of existing OpenNIC domains and DNS services. We deploy a public Tier-2 OpenNIC DNS resolver for over six months and collect over 300 million queries. We continuously monitor zone file changes and employ various open-source databases to identify malicious and suspicious domains. We characterize botnet activities and develop highly accurate machine learning classifiers that can detect new malicious domains and associated botnets. In total, we identify hundreds of potentially malicious OpenNIC domains, and observe that nearly half of the OpenNIC traffic is associated with malicious activity. Our study also provides insights into OpenNIC DNS users and presents several interesting observations.

Index Terms—DNS Security, Botnet, OpenNIC

I. INTRODUCTION

The Domain Name System (DNS) is a crucial component of the Internet, supporting almost all online activities. For years, domain resolution has been primarily performed by public DNS resolvers with root zones managed by ICANN (i.e., TLD delegations such as gTLDs like .com, .org) [1]. However, concerns about privacy and censorship in centralized DNS providers have driven the development of democratically governed DNS [2], [3]. OpenNIC (Open Network Information Center) [4], is a user-controlled alternative to ICANN, offering its own TLDs with community-driven governance. Operated by decentralized volunteers, OpenNIC DNS servers can resolve host names both in the legacy ICANN root and in OpenNIC-operated namespaces, as well as in the namespaces that have peering agreements with OpenNIC [5].

Unfortunately, similar to public DNS resolvers serving the ICANN-managed root, OpenNIC is also vulnerable to network attacks and can be exploited as a springboard for malicious activities. It has been reported that domains under OpenNIC-operated TLDs (i.e., OpenNIC domains) are involved with malicious activities like C2 (command and control) for botnets [6], [7], [8], [9], [10], [11], due to lack of regulation in domain registration. For example, in 2022, the Fodcha botnet utilized 14 OpenNIC domains for C2 servers, resulting in approximately 60,000 active bot nodes per day and launching over 20,000 cumulative attacks targeting over 100 entities [8]. Additionally, CatDDoS, a variant of the Mirai botnet named by its early domain catddos.pirate, spreads via 0-day vulnerability

in IoT devices and employs 22 OpenNIC domains for its C2 servers and targets products of 53 vendors worldwide, with the maximum number of targets exceeding 300 per day in 2024 [6]. Mirai.TBOT, first identified in late 2022 targeting IoT devices, used 93 OpenNIC domains across more than 100 bot groups, with over 30,000 bot IPs per day exploiting 32 vulnerabilities, including two confirmed zero-days [7]. Rap- perBot [11], also discovered in 2022, resurged in Feb/2025, using three OpenNIC domains to attack DeepSeek. The peak number of daily bot IPs reached over 50,000. These incidents show that OpenNIC has been heavily abused by attackers, and the actual scale of botnet activity may be even greater. Although OpenNIC has been operated for more than 20 years, no systematic study has been conducted to understand its behaviors and user activities, especially malicious botnet-related activities.

In this paper, we present the first comprehensive measurement of OpenNIC domains and DNS services, providing a detailed analysis of user activities and potential botnet behaviors. We have deployed a public Tier-2 OpenNIC DNS resolver operating for over six months (from Nov/20/2024 to Jun/15/2025), and listed it on the OpenNIC official website. We have continuously monitored zone file changes (e.g., OpenNIC TLD zone files from OpenNIC Tier-1 servers), recording various information such as domain registration, deletion, and record updates. To focus specifically on OpenNIC-related data, we further set up a hidden DNS resolver with an undisclosed identity to filter out background traffic such as active scans or amplification attacks. During our measurement period, we have collected over 300 million DNS queries. Specifically, our measurement study carefully follows OpenNIC’s privacy policy. We clearly state that our server maintains logs and does not anonymize them to inform users.

Collaborated with Palo Alto Networks, we have carefully inspected each registered domain along with their IPs to identify potential malicious or suspicious activities. We use multiple open-source databases and threat intelligence platforms to check if there are any related IoCs (indicators of compromise) [12], open reports, or malicious files. We have identified over 200 OpenNIC domains that are potentially involved in malicious activities. We further cross-check these domains in our query logs, and summarize several categories of botnets, including different Mirai variants [6], Gafgyt botnet variant [13], and botnets/malware by peered TLD domains (e.g., Namecoin [14] and Emercoin [15]). Using the first three

months of data, we present a detailed characterization of these botnets, and identify four features that distinguish them from benign activities. For example, we find that some OpenNIC botnet domains not only have significantly more records than normal domains, but also are queried by a large number of different IPs. We also observe that source IPs querying potential OpenNIC botnets may only query botnet domains, but with a large number of queries.

To further detect malicious domains and their activities on OpenNIC, we develop a log-based [16] machine learning detector using four identified features. We have tested four classifiers, including Random Forest, Decision Trees, CNN (Convolutional Neural Network), and MLP (Multilayer Perceptron). Results show that all classifiers can achieve very good performances in various settings. By adopting our detector to the rest of query logs (from Feb/28 to Jun/15), we are able to identify several new malicious OpenNIC domains, including four domains without any IoC information from open databases and intelligence platforms. In addition, we have observed a huge breakout of botnet activities starting on May/3, reaching over 70% of daily traffic in our resolver.

Finally, we have analyzed background traffic, including active scans, DNS amplification attacks, and random subdomain attacks. We also present several interesting observations on benign user activities. For example, we find that accessing sensitive content is actually not dominating in benign traffic. We have timely disclosed our findings to OpenNIC administrators, related service providers, and public malware platforms. The OpenNIC administrator team has been working on deleting related domains and establishing a blocklist based on our report. We have also received responses from cloud server operators, indicating that they will investigate our report.

Overall, our contributions are threefold. First, we present the first in-depth measurement study of OpenNIC, characterizing both malicious and benign DNS traffic. Second, we propose a practical solution with new features derived from DNS query characteristics, enabling volunteer operators to continuously monitor and flag potential botnet domains. Third, we identify multiple suspicious and malicious domains and assist OpenNIC operators in taking them down.

Terminology. In this paper, we use *ICANN root* and *ICANN TLDs* to refer to the legacy ICANN-coordinated public DNS root and TLDs managed by ICANN (i.e., TLD delegations such as gTLDs like .com, .org, and ccTLDs like .us). We use *non-ICANN TLDs* or *alternative TLDs* to denote TLDs not present in the ICANN root zone (e.g., .geek, .fur). *ICANN domains* refer to domains under ICANN-managed TLDs (e.g., google.com), and *OpenNIC domains* refer to domains under OpenNIC-specific TLDs (e.g., grep.geek) that are not present in the ICANN root zone.

II. BACKGROUND

A. The OpenNIC Project

Started in 2000 [17], the OpenNIC project [4] is a user-controlled network information center aiming to provide an alternative DNS solution to ICANN. The idea is to introduce

a democratically governed DNS that operates independently of for-profit corporations. Specifically, OpenNIC DNS servers are managed by anonymous volunteers. While OpenNIC supports all ICANN TLDs, it also offers non-ICANN ones for domains that can be registered for free. Table I shows the categories of TLDs that OpenNIC servers can resolve. Currently, there are 16 active OpenNIC TLDs. The TLD .free is inactive and replaced by .libre. OpenNIC has formed peer agreements with three other organizations, Emercoin, FurNIC, and New Nations, to resolve domains under their TLDs. All the peered TLDs of New Nations are inactive now and cannot be resolved because of its service interruption. However, peered TLDs of this category are still listed in OpenNIC community. OpenNIC had also peered with Namecoin for .bit TLD, but terminated the agreement in 2019 due to security issues [18].

B. OpenNIC DNS Resolvers

The infrastructure of OpenNIC DNS service contains three levels of servers, *Root*, *Tier-1*, and *Tier-2*. The root server (e.g., ns0.opennic.glue) is not used to recursively resolve domain names of public users. Instead, they resolve recursive queries from Tier-1 and Tier-2 servers, and provide zone files of glue records and root records to the Tier-1 and Tier-2 servers. OpenNIC root zone contains records for both OpenNIC TLDs and delegation records for ICANN-root TLDs. Thus, Tier-2 servers can fetch them to resolve recursive DNS queries for both OpenNIC and ICANN domains. Tier-1 servers (e.g., ns*.opennic.glue) are the core DNS servers hosting authoritative zones for all OpenNIC TLDs. Each Tier-1 server hosts one or more OpenNIC TLD zones, and Tier-2 servers can update the local copies of zone files from each Tier-1 server anytime. Tier-2 servers (e.g., ns*.location.dns.opennic.glue) are user-operating DNS resolvers open for public use. Anyone can register and operate their own public Tier-2 server. Currently, there are 5 Tier-1 servers and 46 Tier-2 servers located in 13 countries.

Figure 2 illustrates DNS query traces for an OpenNIC user visiting an OpenNIC domain and ICANN domain. Initially, both the root server and the Tier-1 servers transfer the zone files to Tier-2 servers. Each Tier-1 server provides zone files of its hosted OpenNIC TLDs (①), and the root server provides zone files for the glue records and the root [.] records for domains of OpenNIC TLDs, peered TLDs, and ICANN TLDs (②). The Tier-2 server updates these zone files regularly. If any TLD zones, the root zone, or glue records are updated, the Tier-2 servers can periodically retrieve the latest zone records and update their local storage (①②). When the user queries an OpenNIC domain to a Tier-2 server (③), the Tier-2 server looks up the local zone file for the relevant TLD and responds with the record (④). When the user queries an ICANN domain to a Tier-2 server (⑤), the Tier-2 server looks up the local-stored root zone file for a TLD nameserver hosting ICANN TLDs (⑥) and starts recursive lookups (⑦⑧). In step ⑦, the .com TLD nameserver answers the Tier-2 server with the record of the authoritative nameserver that hosts DNS records for the queried domain. In step ⑧, the Tier-2 server then sends

Fig. 1: An example of DNS query log from the client.

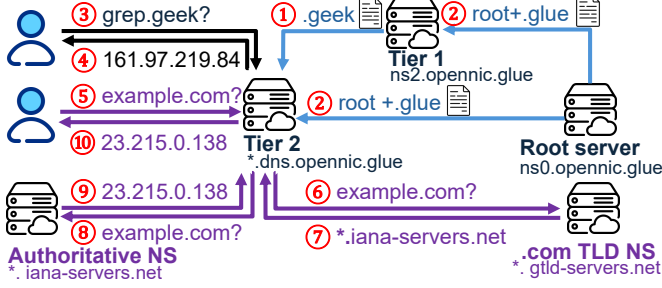


Fig. 2: DNS resolution through OpenNIC resolvers.

a query request to the corresponding authoritative nameserver. After receiving the response from the authoritative nameserver (⑨), the Tier-2 server transfers the answer to the user (⑩).

C. OpenNIC DNS Query and Log

By privacy policy of OpenNIC [19], the administrators of OpenNIC DNS servers are allowed to gather DNS logs. The log function provided by BIND9 [20] collects DNS query records, including timestamp, source IP, source port, query name, query class, query type, query label, and the destination port (since version 9.21.5). For the log data, the *query class* and *query type* refer to the resource records TYPE and CLASS defined in Section 3 of RFC 1035 [21]. The *query label* is the specific label given by BIND9 in the query logs to show the properties of each query. The definition of the *query label* refers to BIND 9 Administrator Reference Manual [22]. An example of the DNS query record in BIND9 log is illustrated in Figure 1. On Mar/21/2025, at 17:55:57.897, a query was sent from the IP address a.b.c.d on port 56396. The client requested A record for domain grep.geek via port 853 for DNS over TLS (DoT) query. The label +E(0)TD means that the Recursion Desired flag (+), EDNS with version number 0 (E(0)), TCP (T), DNSSEC OK (D) were set in this query.

D. Malicious Activities

Botnets. A botnet is a network of compromised devices infected with malware for malicious activities, including distributed denial-of-service (DDoS) attacks and malware dissemination. The Command and Control (C2) servers are used for sending commands or updates to bots. For better spreading of malware, DNS services can be exploited by bots, where the compromised devices (bots) send DNS queries to get the latest IPs of C2 servers [23].

Techniques like Fast Flux and Domain Generation Algorithms (DGAs) are commonly used for botnets. Fast flux assigns multiple IPs to the same domain name, and updates IPs at high frequency. DNS Fast Flux [24], [25] turns malicious domains into shifting targets to evade detection. DGA generates a large number of new domain names, in the form of either random strings (e.g., cehhimcjhio.bazar),

Category	TLDs	Status
ICANN	.	Active
OpenNIC	.bbs, .chan, .cyb, .dyn, .epic, .geek, .gopher, .indy, .libre, .neo, .null, .o, .oss, .oz, .parody, .pirate	Active
OpenNIC	.free	Inactive
Emercoin	.bazar, .coin, .emc, .lib	Active
FurNIC	.fur	Active
New Nations	.ku, .te, .ti, .uu, .ko, .rm	Inactive
Namecoin	.bit	Terminated

TABLE I: TLDs that OpenNIC servers resolve or peer.

or by concatenating several words together for a potentially meaningful phrase (e.g., tryanotherunique.club). Some DGAs are time-based, generating different domains for each time period, and some include seed values to make it more difficult to predict future domains [26]. According to the open-source database DGArchive [27], there are 146 malware families using DGAs for domain generation.

DNS Amplification Attack is a form of DDoS [28], where the attacker uses a public DNS server for flooding DNS lookups with the return addresses spoofed to the victim's addresses. The amplification factor represents the ratio of the response size to the request size [29]. For example, sending one ANY query to the authoritative name server can yield a response with an amplification factor as high as 98.3 [29]. The flooding response to the victims can overwhelm their server and network infrastructure.

Pseudo-Random Subdomain Attacks, also known as random prefix attacks, is a type of DDoS where the attacker continuously sends queries for subdomains with random strings (e.g., 4Vg6pA8.victim.com). Due to the non-existence of records for random subdomains, the authoritative server will respond NXDOMAIN to the resolver. This attack can increase the burden on both the queried resolver and its authoritative server, declining its service quality [30].

E. Motivation of Our Study

OpenNIC was established as a decentralized, non-profit, user-governed DNS network, offering alternative DNS services free from political or commercial control. It has been operational for over 20 years, attracting users who seek alternatives for privacy, governance, or freedom from ISP-imposed restrictions. Prior discussions [31], [32], [33], [34] indicate that OpenNIC provides value by moving DNS resolution away from ISP-controlled resolvers, helping users avoid potential censorship. Additionally, OpenNIC supports TLDs and domain registration outside ICANN oversight, appealing to communities seeking uncensored or unconventional naming services. However, OpenNIC's open architecture also makes it susceptible to abuse, particularly by botnets. Recent reports highlight widespread misuse of OpenNIC services for malicious campaigns [6], [7], [8], [9], [10], [11].

Despite its long history and potential security implications, user behavior on OpenNIC remains largely unexplored. Understanding both benign and malicious activities is critical for improving DNS security and Internet resilience. In this work,

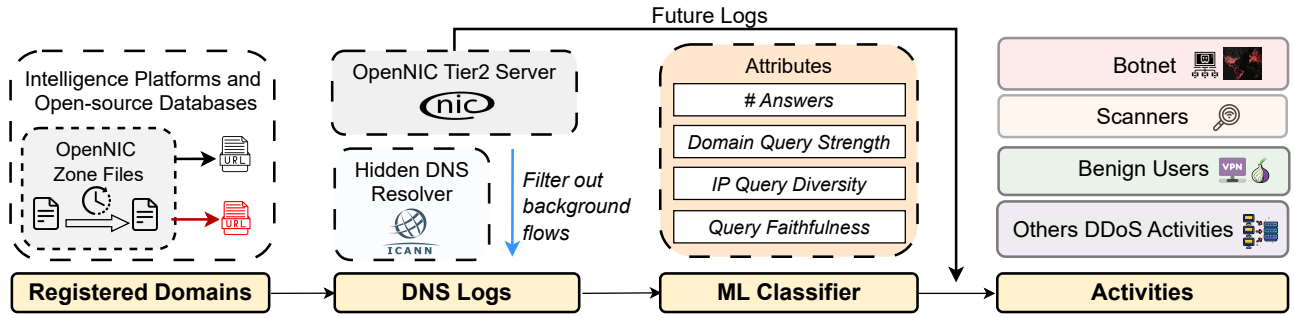


Fig. 3: Illustration of malicious domain detection and query classifier framework.

we aim to fill this gap by providing a comprehensive assessment of OpenNIC usage. Specifically, we investigate: (1) the extent to which OpenNIC services are exploited for malicious purposes; (2) what legitimate users do with OpenNIC-resolved domains; and (3) whether patterns in OpenNIC activity can inform data-driven strategies to proactively mitigate abuse.

III. RESEARCH OVERVIEW

In this paper, we present a measurement study on OpenNIC domains and DNS services to investigate the user activities, especially malicious botnet-related activities.

A. Research Methodology

Figure 3 illustrates the workflow of our research. We start with OpenNIC domain analysis, where we continuously monitor zone file changes of each TLD, and attempt to identify potential malicious domains. Specifically, we utilize open-source intelligence platforms and databases to identify reports of malicious activities related to these domains or their resolved IPs (Section III-D). Additionally, we set up a Tier-2 OpenNIC DNS resolver, publish it to the OpenNIC public server list [35], and collect DNS query logs. Meanwhile, we set up a hidden DNS resolver to filter out background noise on our OpenNIC server (Section III-B). Based on our data collection on a 90-day period, we summarize main botnet activities and four characteristics in malicious OpenNIC domains (Section IV). With these features, we implement machine learning classifiers that can achieve high accuracy, adopt them to the rest of data logs, and report meaningful discoveries (Section V). Finally, we summarize other user activities, including those of benign users and other DDoS activities in queries (Section VI).

B. Data Collection

OpenNIC allows any anonymous volunteer to operate a Tier-2 server. In our experiments, we set up a Tier-2 server as an OpenNIC DNS resolver [36]. To make our resolver more attractive, we provide not only the traditional DNS service on port 53 with the support of both UDP and TCP, but also DNS over TLS (DoT) on port 853 and DNS over HTTPS (DoH) on port 443. Besides the log function provided by BIND9 (Section II-C), we also use `DNSTAP` [37] to record DNS query answers. In addition, our OpenNIC resolver automatically updates the zone files for each TLD every 50 minutes. We collect all changes during each update and record the differences

Category	URLs	Category	URLs
Parked or Blank	176	Education and Tools	5
Personal Sites and Blogs	40	Games	5
Login Portals	15	Religion	5
Business and Economy	8	Piracy	3
Dynamic DNS	7	Others	54

TABLE II: The categories of websites using OpenNIC TLDs.

between two consecutive versions of the zone files, including domain registrations, deletions, and DNS record updates. Our resolver is published on the OpenNIC public servers [35].

Our primary objective of data collection is to obtain all DNS queries that are specifically targeting the OpenNIC infrastructure. However, our Tier-2 server inevitably records background network traffic, including random IP scanning and other traffic targeting all DNS resolvers (e.g., DNS amplification attacks).

To systematically filter out such background noise from our data collection, we deploy a non-public DNS resolver exclusively for resolving ICANN domains. We ensure that the IP address and other information of the hidden resolver remain undisclosed. By recording and analyzing the traffic captured by the hidden resolver, we are able to identify and characterize background noise traffic and effectively remove requests with similar patterns from our OpenNIC resolver (Section VI).

C. Domain Dataset

The zone files collected from our Tier-2 resolver contain a total of 1,812 domains registered under OpenNIC TLDs. Among them, 55 domains are dedicated to OpenNIC services and community websites, while 69 domains are associated with private or local IP addresses as indicated by their A records. We issue HTTP and HTTPS requests to all of them. The results show that only 318 (18.8%) of 1,688 domains successfully resolve to a valid web server. We collect HTML content from the homepages of the 318 accessible domains and manually check them for categorization. Table II presents a breakdown of the top categories of these websites. 176 domains are parked or showing blank pages. Among the remaining 142 domains, the majority are personal websites and blogs (40), followed by login portals (15), business (8), dynamic DNS (7), education (5), games (5), religion (5), and piracy (3). Among all the accessible websites, 34 of them contain non-English content, including Russian (7), Japanese (6), Chinese (4), German (4), and others.

D. Identifying Potentially Malicious Domains

We evaluate all OpenNIC domains and their A records using the VirusTotal API [38], a publicly available service for evaluating potentially malicious files, IP addresses, and URLs. A domain is classified as potentially malicious if (1) the VirusTotal detection score for the domain is not 0; or (2) if VirusTotal flags malicious files under the *Communicating Files* section of the *Domain Relations* tab, and these files are linked to the domain (i.e., the domain is queried when the files are opened or executed).

A domain is considered suspicious and requires further investigation if: (3) the domain resolves to IP addresses that are marked as malicious, or (4) the resolved IP addresses are associated with malicious communicating files or other malicious domains. According to the *Passive DNS Replication* in VirusTotal *IP Relations* tab, additional malicious domains with the same A records can be identified.

One feature of malicious domains is that multiple resolved IP addresses of their domains might be frequently updated [23], [39] (e.g., Fast Flux [40]). To capture this, we also keep monitoring the changes of zone files, and we identify additional suspicious domains based on the following heuristic characteristics: (5) if there are two updates to the domain's DNS records within one day, or (6) if there are more than three A records for the domain after any updates. Some OpenNIC domains only have TXT records, which can also be used for malicious activities [41]. We also examine these domains.

To confirm malicious domains, we further utilize several open source tools and databases, including Rösti [42], MalwareBazaar [43], ThreatFox [44], JexBoss [45], ELF DIGEST [46], AnyRun [47], and QAX Threat Intelligence Center [48]. These sources have been widely used in previous research work [49], [50], [51], [52], [53] or industry practice [54], [55], [56], [57], demonstrating their reliability and broad acceptance. They help us check for any existing indicators of compromise (IoC) or reports related to malicious files associated with these domains.

In total, zone file analysis reveals 239 malicious domains by Jun/18/2025. Among these domains, 50 have only TXT records for IP addresses either in plaintext, encoded, or encrypted. Additionally, 10 domains are marked suspicious. Among them, four domains are confirmed malicious as discussed in Section V-C; three domains have IPs for web hosting services as A records; two domains with the same A record labeled as *BOT* by QAX; and one domain labeled as *Criminal IP* by VirusTotal but actually hosting a valid website. Among the confirmed malicious domains, we observe domain squatting [58] targeting well-known brands or official OpenNIC services (e.g., hikvision.geek for network video recorder malware spreading [59]; opennicdemon.geek for HailBot [60] botnet C2) and some domain names containing discriminatory words.

Not all malicious domains are captured in our DNS query logs. Specifically, 25 registered malicious domains, including four domains marked suspicious but confirmed malicious (in Section V-C) appear for botnet activities. In this study, we

focus on these malicious domains appearing in logs.

E. DNS Query Datasets

Our data collection lasts 208 days, from Nov/20/2024 to Jun/15/2025. In total, we have collected 313,922,217 DNS queries from our Tier-2 OpenNIC resolver. These DNS queries can be categorized into three main types: (1) liveness checks by the OpenNIC community or from the OpenNIC Wizard DNS client application [61]; (2) background noise showing patterns similar to traffic collected by our hidden resolver; (3) OpenNIC traffic of interest for our study.

Liveness checks by the OpenNIC community are periodically sent from higher-tier OpenNIC resolvers every 15 minutes. These resolvers have verifiable static IP addresses. We identify 2,218,764 such DNS queries and remove them from our dataset. For the liveness checks conducted by the OpenNIC Wizard DNS client, queries for 14 domains from a fixed list are performed periodically. We remove 2,307,274 such liveness check DNS queries. The remaining 309,396,179 queries are used as the total number of queries.

Next, we leverage our hidden DNS resolver to filter background traffic. In total, we record 11,805,497 DNS queries from our hidden resolver. Among these, 23,783 belong to active Internet measurement scans, and the remaining are mainly random prefix attacks and DNS amplification attacks. We use source IP addresses, queried domains, and other relevant features to filter traffic with similar patterns. Section VI presents more details.

After removing the unwanted DNS traffic, our dataset contains 300,049,431 remaining DNS logs. Among these queries, 61,244,686 are from benign users, while 238,804,745 are potential malicious activities or botnets that attempt to exploit OpenNIC infrastructure.

Figure 4 shows the traffic distribution and the percentage of queries from benign users since Dec/1/2024. Before Jan/11/2025, most queries are from non-benign activities. After that, benign traffic consistently increases every day, and gradually reaches over 50% of the total traffic, indicating that our server is gradually configured by more users. Since April, benign queries have remained stable. However, we observe a breakout of botnet queries on May/3/2025 (details in Section V-C), which reduces the proportion of benign queries to about 13.7%. The *Others* indicates queries whose patterns are similar to background servers or active scans (Section VI).

F. Ethic Considerations

The Institutional Review Board (IRB) at our institute has determined that our research *does not meet the definition of human subject research*. We also carefully reviewed and followed the privacy policy of OpenNIC, which mentions that Tier-2 servers are not covered by this policy [19]. Also, Tier-2 DNS servers are permitted to retain non-anonymized logs, and currently *there is no official OpenNIC policy that addresses the privacy and retention of named logs* [62].

We have further communicated with the OpenNIC administrator and informed them of our work. We notified them that

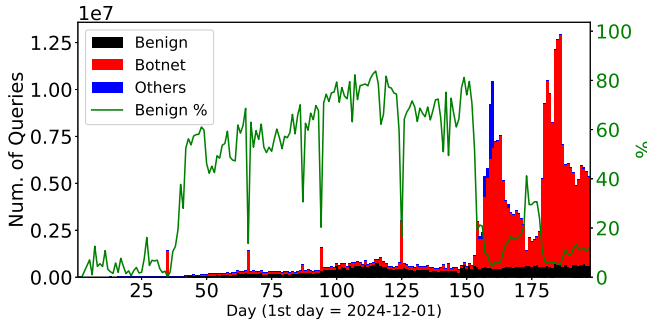


Fig. 4: Daily Query Counts and Benign Traffic %

our server retained the logs including source IPs, and they did not express any objection. To mitigate the impact of botnets and other cyberattacks, we have timely blocked outgoing network flows for DNS query responses for botnet domains, TXT queries for potential amplification attack domains, and all ANY queries (which can be abused for amplification attacks).

Informed consent. In line with the principle of informed consent, on the OpenNIC server listing website, we clearly state that our server will keep logs and will not anonymize logs. If the benign user actively chooses our server for DNS resolving, they should be notified of our log policy. Thus, if they decide to opt-out of our resolver, they can easily remove our DNS server from their network settings.

Responsible disclosure. We provided all confirmed malicious and suspicious domains, with their DNS records and related reference or IoC reports, to the OpenNIC community via its official IRC channel. Based on the feedback, OpenNIC administrators have started re-verifying and taking down malicious domains. They are also developing an automated monitoring system to identify and take down potentially malicious domains based on the latest IoCs from open-source databases and threat intelligence platforms, with our suggestions.

For the malicious domains hosted by other dynamic DNS services or domain registrars, we have contacted the service administrators, but only one service provider responded to our request. For the IPs involved in malicious activities according to our findings, we also contacted the cloud server operators hosting these IPs with the related public IoCs. We received only one follow-up response indicating that they will investigate our report.

IV. BOTNET ACTIVITIES IN OPENNIC

In this section, we examine and analyze botnet activities within the OpenNIC ecosystem. We choose queries from the first three months during Dec/1/2024 and Feb/28/2025 for preliminary analysis. This dataset allows us to identify several categories of botnets observed in our measurements and to highlight the statistical differences between malicious and benign queries. Based on these differences, we further extract four key attributes, which are later used for classification.

A. Botnet Domain Categorization

We carefully examine DNS queries for the 239 identified malicious domains. In total, during the first three months we

identify 13 OpenNIC domains used for botnets. We also see 33 malicious OpenNIC domains appear in query logs but have been deleted before our server began operating, 321 domains from the current peered service Emercoin, and 3 domains from the terminated peered service Namecoin. We categorize malicious domains that are queried by a large number of users based on malware types.

Category 1: Mirai Variant - HailBot. We find that nine OpenNIC domains, brandlane.parody, nextstep.dyn, craftsui.te.gopher, buildbetter.oss, securevault.pirate, clearpath.libre, adaptiveflow.dyn, insight.geek, and localmarket.indy have appeared in the query logs since Dec/5/2024. These domains share the exact same A records (initially 15 A records and becoming 27 A records on Feb/25/2025). By IP reverse lookup, himanurnice.ru and iholdxrp.ru also have the exact same A records with these OpenNIC domains, and the former also appears in the query logs. According to IoC of himanurnice.ru published on Feb/1/2025 [63], all these domains with the same binding IPs are used as C2 servers for Mirai botnet with HailBot tag with multiple references [60], [64], [65], [66]. Hailbot, named by the string output to the terminal, is a Mirai variant first discovered in Feb/23/2024, and spreads by exploiting UPnP vulnerabilities of certain vendors [67]. During the first three months, 4,038,058 queries from 200,045 IPs in 149 countries have queried these malicious domains, which are potential victims. Most victim IPs are in India (31.46%) and China (22.81%). Particularly, it surged on Jan/4/2025 and Feb/4/2025, with more than 14,600 newly appeared source IPs.

Category 2: Mirai Variant - CatDDoS. We discover that we lovedky.oss shares the same A record with a malicious domain banthis.su according to VirusTotal. The domain banthis.su is identified to be used for CatDDoS botnet [68], [69], [70], which is a variant of the Mirai botnet since August 2023 [6]. It experienced stable growth from Dec/11/2024 to Feb/19/2025, and then started to decline. During the first three months, victims are 140,772 IPs in 86 countries, most of which are in India (54.86%) and China (36.99%). There are 48,283 IPs that are also victims of the Mirai variant of Category 1.

Category 3: Gafgyt Botnet. Gafgyt [13] is an active botnet malware targeting IoT devices since 2014. One particular interesting finding is a group of 2 OpenNIC domains (trump2024.oss, liberalretard.libre), which experienced a concentrated outbreak since Jan/19/2025 (with 2,045 newly appeared source IPs), one day ahead of the Inauguration Day of the United States. There are 19 A records for these domains when they initially appeared in queries. By DNS reverse results and existing reports [71], [72], xaiverbot.net shares the same A records with them, with the malware label as Gafgyt [71]. Also, on Feb/23/2025, domain blinkercentral.oss was registered with 35 A records, all of which were previously resolved to xaiverbot.net. During the first three months, 10,570 source IPs from 112 countries have queried the above domains with 1,165,093 queries. Most victim IPs are in Taiwan (14.48%), Hong Kong (12.40%), and the United States (9.12%).

Category 4: Mirai/Pupy. We find that nine IPs keep querying the domain helloworld.libre, which had been deleted before

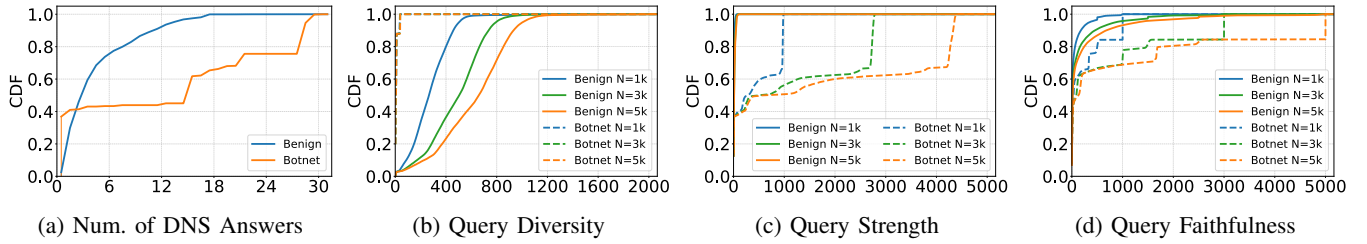


Fig. 5: CDF of selected features for botnet and benign queries.

our server started operating. There are 1,330,597 queries sent until Feb/28/2025. According to IoC by QAX platform, the relevant malicious files are classified as malware of Mirai botnet and Pupy Trojan. Pupy [73] is a cross-platform RAT and post-exploitation framework. It can migrate into processes by reflective injection and load remote codes and Python libraries from memory.

Category 5: Botnet and Malware by peered TLD domains.

The .bit TLD is based on Namecoin [14], which leverages blockchain to decentralize website addresses. OpenNIC terminated its support in 2019 due to extensive malware spread across it [18]. However, these domains are still queried by victims currently, with the possible reason that the OpenNIC API was used for malware development to return the nearest OpenNIC server IPs. Three malicious domains with a .bit TLD (i.e., athery.bit, babloom.bit, and floppys.bit) were queried by 76 IPs in United States and Cayman Islands. These domains are reportedly used as C2 servers by PyXie, a Python-based Remote Access Trojan (RAT) discovered in 2019 [74], [75].

In addition, the .bazar TLD is managed by Emercoin, a blockchain platform that provides non-ICANN TLDs and domain registration [15]. Since Jan/9/2025, there have been 494 queries from 5 source IPs for 325 domains with the .bazar TLD. DGArchive [27] confirms 28 domains belonging to the BazarBackdoor DGA malware family (a.k.a Bazarloader), which is a Windows-based backdoor using Emercoin domains (.bazar) for malware spread. 213 domains are suspected to belong to BazarBackdoor by regex guesses.

B. OpenNIC Botnets Characterization

Based on identified botnet domains and their activities, we attempt to summarize the statistical differences that can set them apart. Our preliminary analysis contains 8,391,088 botnet queries and 7,389,543 benign queries in total. We aim to find some features that can be used to develop a classifier to detect incoming botnet queries.

Number of DNS answers (Ans.) We find that some botnet domains have significantly more records than normal domains. For example, for botnet domains in Mirai and Gafgyt categories, the number of A records for each domain can be up to 28. For comparison, 90% of normal domain queries return fewer than 10 records. Moreover, for some botnet domains using TLDs of peered networks like .bit, or some domains that have been deregistered, the number of A records is zero. Therefore, the number of answers in the query response can be a potential feature to distinguish whether the queried domain is used for a botnet or not. The Cumulative Distribution Function

(CDF) of answer numbers for both potential botnet and non-botnet domains, as illustrated in Figure 5a, reveals significant differences between them.

Query diversity of IP (Dvs.) We also observe that source IPs querying potential OpenNIC botnet domains typically only query botnet domains, in contrast to benign users who query a wider variety of domains. For example, on Feb/4/2025, the IPs querying Mirai botnet domains contacted fewer than 10 distinct domains, whereas benign users typically query over 1,000 distinct domains within 5,000 queries. To capture this, we define a value of *Query Diversity* for each query in logs. The value is the number of distinct queried domains in the last N queries by the source IP. We empirically select N as 1000, 3000, and 5000. The CDF in Figure 5b shows that it is easy to distinguish the differences for all three N values.

Query strength of domain (Str.) Botnet domains are typically queried by a large number of different source IPs [76], making queries of these domains distinct from other benign ones. We also observe this pattern in our OpenNIC logs: some OpenNIC botnet domains are queried by larger numbers of users than others. We define a *Query Strength* of the domain in each query as the number of distinct source IPs querying the current domain in the last N queries. Figure 5c shows the CDF of values when N=1000, 3000, and 5000. We can also see a visual distinction between the distribution of benign users and botnet victims. With a higher N value, the query strength of botnet queries is further distanced from benign queries.

Query faithfulness (Fth.) We observe that some botnets are only queried by a small set of IPs, and these victim IPs only query these botnets. For example, only a limited number of IPs are victims of botnets using .bit domains, and they only send queries to these botnets. To capture this observation, we define a *Query Faithfulness* of each query as the number of queries for the same domains in the last N queries from the same source IP. As Figure 5d shows, this feature helps identify botnet domains that are queried by only a small number of victims, while a higher N value provides a clearer distinction between the distributions of botnet and non-botnet queries.

Example. Table III shows a snapshot of DNS queries in March within one second, where malicious domain queries are labeled in grey. The feature values are based on N=5000. The *Query Diversity* (Dvs.) values for malicious domain queries are significantly lower than those for benign queries. The malicious domain blinkercentral.oss has 35 DNS answers (Ans.), but others may have zero answers due to being previously taken down. Queries for domains blinkercentral.oss and

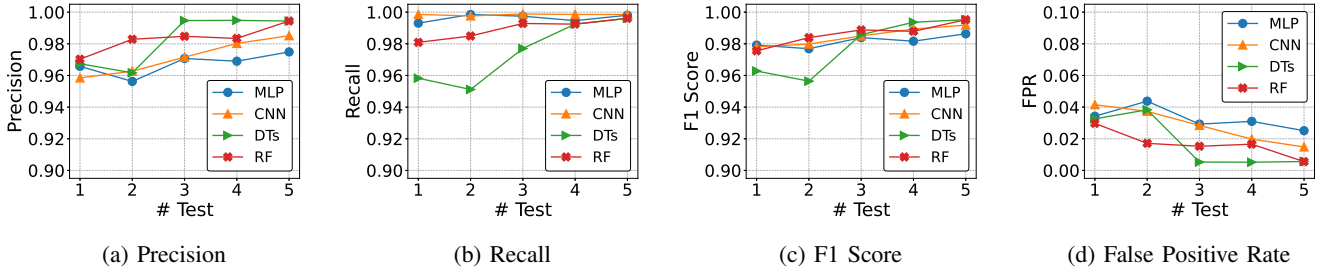


Fig. 6: Prediction metrics of each test (labeled by # Test).

Src IP	Domain	Ans.	Dvs.	Str.	Fth.
a	*.spotify.com	2	347	7	136
T	helloworld.libre	0	1	6	5000
b	*.google.com	1	913	14	15
b	*.google.com	0	913	14	16
b	*.google.com	1	913	14	17
U	blinkercentral.oss	35	2	1822	1
V	floppy.bit	0	3	16	1640
W	projectbluebeam.dyn	0	2	1836	12
c	netgear.com	9	350	4	124
d	netgear.com	9	347	4	124
X	babloom.bit	0	3	16	1665
e	*.ip6.arpa	0	351	3	1
Y	blinkercentral.oss	35	2	1821	7
Z	projectbluebeam.dyn	0	2	1835	5
f	lut*.com	1	501	4	230

TABLE III: DNS query fragment with characterization values.

projectbluebeam.dyn have significantly higher *Query Strength* (Str.) values, but values for other malicious domain queries helloworld.libre and *.bit are similar to benign ones. However, the significantly higher *Query Faithfulness* (Fth.) values can distinguish them from benign queries.

V. DETECTING OPENNIC BOTNET DOMAINS

Based on the botnet characteristics in OpenNIC, we aim to develop machine learning based classifiers to actively detect potential botnet queries. The goal is to capture undiscovered malicious domains used by widely spread botnets according to query behaviors.

A. Implementation Setups

We label all botnet domain queries in the query logs according to previous malicious domain detection results in Section III-D, and choose $N=5000$ for the features mentioned in Section IV. Binary labels are assigned to each query as either botnet-related or not. We remove all hidden-resolver traffic observed at our Tier-2 server. Based on identified characteristics (introduced in Section VI), we further filter similar traffic patterns. For example, we remove all ANY queries for DNS amplification attacks, which are out of scope for botnet domain detection. During the initial days of server operation, there are not enough queries from benign users. Thus, we omit these data and select logs since Jan/25/2025.

To demonstrate the effectiveness of our features, we conduct the experiment using three setups. First, we use logs in January (starting from Jan/25/2025) and February for training to predict data in March and April (Test 1). Next, since

characteristics are potentially time-varying, we use the logs from Jan/25 to February, along with data from the first week (Test 2), half month (Test 3), three weeks (Test 4), and the full month of March (Test 5) for training, and predict the remaining logs until April. We analyze the performance metrics and use correlation matrices to illustrate relations between multiple variables and between variables and the label. Finally, we apply our classifier to rest logs to identify any new findings.

Algorithms. We select four machine learning classifier algorithms, Random Forest (RF), Multi-Layer Perceptron (MLP), Decision Trees (DT), and Convolutional Neural Network (CNN). More specifically, the scikit-learn [77] is used for RF and DTs implementation with `n_estimators` set as 25 for RF and 15 random states for both. MLP and CNN are implemented by PyTorch [78]. The MLP structure contains three fully connected layers, with 64 and 32 hidden layer nodes, respectively, and a ReLU activation function after each hidden layer. To adapt to the input features, the network removes redundant single-channel dimensions before the forward propagation begins. This CNN model applies two convolutional layers with 16 and 32 filters, respectively, with a kernel size of 3 and padding to preserve the signal length. The output is passed through a fully connected layer of 64 units with ReLU activation and 0.3 dropout rate.

B. Evaluation

Performance evaluation. We concentrate on the precision, recall, and false positive rate (FPR) when evaluating the performance. As illustrated in Figure 6, all four algorithms perform well with precision, recall, and F1 score exceeding 95.1% and FPR less than 4.4%. Also, with more recent training data, the performance metrics show an overall increasing trend. In Test 5, when using data prior to April to predict data for April, the RF model achieved over 99.4% in precision, recall, and F1 score, with a false positive rate of merely 0.55%. The results show the effectiveness of our features.

We further check all false-positives. We find that some non-botnet domains are classified as botnet activities. One reason for false positives is that these misclassified queries are sent at higher-than-normal frequencies. For example, on Mar/25, one IP 74.208.a.b (anonymized) kept sending the same query multiple times, with an average frequency of 102.17 queries per second. For 265,341 queries from this IP, 69,955 (26.34%) are predicted as botnet queries by at least

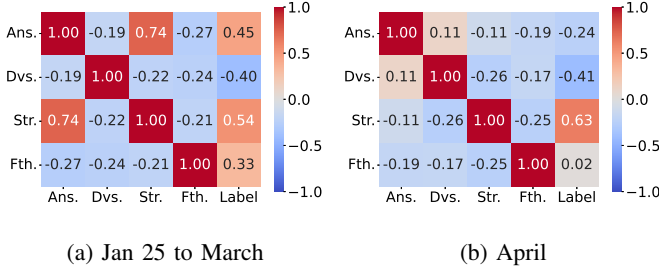


Fig. 7: Correlation matrix.

one of the four models, which affects the precision. The precision also dropped heavily on April/30. One IP `76.90.c.d` (anonymized) continuously queried a live testing URL of a music service, and 12,994 out of 13,002 queries (99.94%) were predicted as botnet queries by at least one of the four models. Also, we find that 51,874 of 145,633 (35.62%) queries for botnet domains are predicted as benign queries, affecting the recall values of RF and DT. However, this does not affect the overall botnet domain prediction, as these domains are identified as botnets by the remaining queries.

Feature correlations. To illustrate the correlations among features and the results, we use the Pearson Correlation Coefficient to show the relationships between features, and the Point-biserial Correlation Coefficient to demonstrate the relationships between features and the results. Figure 7 shows the correlation matrix using labeled data from Jan/25 to April. Overall, both *Query Strength* and *Query Diversity* contribute consistently to the performance. In April, we find that the correlation of DNS answer numbers to the results shifts from positive to negative. Also, for the correlation between features *Query Strength* and *Number of DNS Answers* changes from strong positive to weak negative. Due to the takedown of botnet domains, the number of their A records dropped from significant numbers to zero. The analysis of correlation matrices shows that, despite changes in botnet domain DNS records, our prediction models, utilizing the selected features, continue to perform robustly, achieving high F1 scores and low false positive rates.

Comparison with previous works. We compare our work with several previous works [79], [80], [81], [82] in botnet detection. For comparison, we use the results of RF in Test 5 as a representative of our work. As shown in Table IV, our work performs as well as or better than previous work with fewer features. Previous methods typically can achieve precision values of 95.7%-99.9%. Some prior approaches rely on domain-name features (e.g., length or entropy), which might be more easily evaded, or on features beyond DNS queries that require complex system configurations. Our approach can achieve similar or better results with only 4 features utilized with lightweight deployment.

C. Further Detection

New malicious domains for categorized botnets. We further adopt our ML classifier to analyze the rest of data after Feb/28. We find that additional domains are involved in potentially

Method	#Features	Prec.%	F1 %	Source
Our work	4	99.44	99.53	DNS query
SVM [79]	15	95.7	94.5	PCAP traffic
DL [80]	21	99.99	99.99	NIDS
XAI [81]	3/4/7	/	96.7-99.9	XAI IDS
UMUDGA [82]	29	98.9	98.9	DGA domains

TABLE IV: The comparison of detection effectiveness between our work and other previous works

malicious activities. For example, since Mar/24, the domain `projectbluebeam.dyn` has been queried by the same source IPs of victims of Category 3 (Gafgyt Botnet). There are no published IoCs for this domain until our report. This shows that relying on open-source databases and IoCs for malicious domain detection cannot fully identify newly appearing botnet domains in time. The botnet remained active until May/3 with 1,887,715 queries. Another domain `mineplex.libre` with encoded/encrypted TXT records was registered on Mar/9 and has been queried by victims until May/6 with 5,838 queries. According to ELF DIGEST, the malware is also Gafgyt [83]. Additionally, another terminated peered TLD domain, `checkb ox.bit`, which has been queried continuously since Mar/7, is identified for the CoreBot Banking Trojan [84].

New botnets. Since Mar/3, we observe a group of domains, including `thisisadomain.net`, `mydomain.oss`, `cookie.pirate`, `original.gopher`, `nerd.parody`, and `domain.parody`, are queried. These domains are linked to the same Trojan file [85] and share the same IPs (except `nerd.parody`). There is an interesting finding that the real IPs for C2 servers should be the reverse of IPs in their A records. For example, the initial A records are `226.x.5.230`, which are IP segments for multicast. The reversed forms are like `5.230.226.x`. By checking VirusTotal and QAX, there is no reported IoC for the IPs of A records, but two of the reversed IPs are labeled as malicious. The IoCs of the linked Trojan files by VirusTotal further confirm our inference [85], [86]. Such domains cannot be confirmed through zone file checking described in Section III-D, but can be identified using our ML classifier, demonstrating the effectiveness of our approach.

The breakout. On May/3, we observe a breakout of botnet queries, including 5 OpenNIC domains `bot.pirate`, `devineintell ect.geek`, `iloveto.geek`, `nigga.dyn`, `weloveto.geek`, and 11 non-OpenNIC domains `[boatz/fuckmylife/ge0rgedr0yd/georged r0yd/georgedroyd/ilovefent/jewsdid911/sosa].anondns.net`, `no tagain.gotdns.ch`, `uwukittykats.space`, `kittyuwobotnet.online`. According to IoC reports, VirusTotal, and QAX, this botnet belongs to Mirai and its variations. All these domains are resolved to the same IP. The daily queries of this botnet reach 70% of the total on average since it emerged. We find that 191,631,728 (61.85%) queries were sent from 23,377 victims. From Figure 4, we can observe several surges. For example, two domains were registered on May/6-7, leading to 6,819,404 queries on May/12. Additionally, five domains were registered on May/29-30, resulting in over 8 million queries each day. Finally, two new domains were registered on Jun/2-3, causing over 10 million queries. The large number of queries even

caused the blockage of our server.

We have timely reported our findings to OpenNIC when the first botnet domain appeared on May/3, and these OpenNIC domains were deleted later. The OpenNIC domains created later were auto-deleted from OpenNIC zones. For this period, RF gives the best performance with 99.98% precision, 99.36% recall, 99.67% F1 score, and 0.01% FPR.

VI. OTHER ACTIVITIES

In this section, we first focus on non-botnet but security-relevant traffic patterns, which account for only a small portion of the total queries. We then discuss our observations from the remaining traffic, which we infer to be benign queries from legitimate users.

A. DNS Amplification Attack

Amplification attacks are the most common traffic that appears in our hidden resolver (which also appears in our OpenNIC server), with ANY and TXT query types. In total, 9,139,884 (2.95%) queries are amplification attacks abusing 174 domain names, where 146 domains are queried with ANY type and 28 with TXT type. For example, since May/3, there are 6,451,510 queries of *diux.mil* with query type ANY. The response with 4 DNSKEY and 17 RRSIG makes a total of 3,892 bytes, leading to an amplification factor of 90.5. Among queries for these domains, *reklamco.se* with type ANY produces the largest amplification factor (966.1), with an answer packet of 41,544 bytes containing 49 DNSKEY and 344 RRSIG records. Also, 158 of 174 domains used for amplification attack in OpenNIC server also appeared in the hidden resolver in similar periods. This indicates that some attackers might simply randomly scan the entire IP segment for DNS resolver discovery, instead of specifically choosing our OpenNIC server.

We observe some interesting findings in amplification attack queries. We find that TXT records of benign domains may be abused by attack initiators. For example, TXT records for *sqs.us-east-1.amazonaws.com*, one Amazon service endpoint, yield a single DNS answer of 9,520 bytes, listing IP addresses. TXT records for *coursera.org* make a single DNS answer of 2,755 bytes, and most of them are strings for domain verification by third-party services. Also, we find a group of domains that might be specifically registered for launching amplification attacks. These domains are in the format of *<word>-<word>.ru*. All these domains have TXT records as *google-site-verification=<string>*. For example, *maono-discount.ru* has 507 TXT records with 41,113 bytes query answer. In addition, all these domains host fake shopping websites with the exact same UI template with IPs 188.xxx.xxx.[146-155].

B. Random Subdomain Attacks

Both our servers receive DNS queries that exhibit patterns of random subdomain attacks: queries for numerous non-existent pseudorandom subdomains of the same primary domain are sent to the resolvers. We find that the primary domains are

typically not well-known, and query types vary widely. For example, queries from all IPs in both our OpenNIC DNS server and our hidden DNS resolver contain 24 query types (A, A6, AAAA, CAA, CERT, CNAME, DNAME, DNSKEY, DS, HINFO, HTTPS, LOC, MX, NAPTR, NS, PTR, SMIMEA, SOA, SRV, SSHFP, SVCB, TLSA, TXT, URI) including some less common types. Overall, we find that 93 IPs under 17 /16-subnets are source IPs of random subdomain attacks with 185,526 queries, and only two IPs do not appear in our hidden resolver. In addition, subdomains queried for these attacks are based on 23 second-level domains, which are either parked, hosting a blank page, or hosting a webpage that is not frequently visited.

The standard deviation of the number of DNS query types from a source IP can indicate whether the IP is generating random subdomain queries. Formally, for a source IP i , let $\mathcal{T}_i$ denote the set of distinct DNS query types queried by i in our logs, and let m_i denote the number of distinct DNS query types in queries from i (i.e., $m_i = |\mathcal{T}_i|$). For each type $t \in \mathcal{T}_i$, let $c_{i,t}$ be the number of queries of type t from IP i . We compute the mean count per type as $\mu_i = \frac{1}{m_i} \sum_{t \in \mathcal{T}_i} c_{i,t}$, and the standard deviation of query-type counts as $\sigma_i = \sqrt{\frac{1}{m_i} \sum_{t \in \mathcal{T}_i} (c_{i,t} - \mu_i)^2}$. For IPs issuing random subdomain attack queries, the standard deviation ranges from 1.80 to 19.77, with query counts exceeding 127. In contrast, other IPs show standard deviation values above 54 when they send more than 100 queries. Moreover, IPs performing random subdomain attacks exhibit up to 24 DNS query types, whereas other IPs use no more than 14 types.

C. Active Scans

There are active liveness scans conducted for internet measurement by network/cybersecurity service providers, research institutes, and similar organizations. We examine all the source IPs after removing non-benign activities (e.g., amplification attacks), and further check them in AbuseIPDB [87], which provides the domain, hostname, usage type, and ISP information of IPs. 1,693 IPs are confirmed for active liveness scans during the entire period of server operation with 21,338 queries. There are some unique features for these active liveness scans: (1) some queries are in CH class with TXT type; (2) some query initiators include keywords like *research*, *measurement*, *scan*, *dns*, or URLs of their institutes; (3) some query names are domains of common websites like *google.com*, *example.com*, etc, or root domains like root (.). These queries are only a small part of the entire query log.

D. Benign User Activity

Excluding all malicious and scanning activities in OpenNIC, we infer that the remaining 61,244,686 (19.79%) DNS queries are sent from benign users, as there is no evidence indicating those DNS queries are malicious. During the server's operation, we observed 5,117 unique IPs for benign user activities. There are only 3,098 non-botnet queries for OpenNIC TLD domains from 121 IPs. Excluding OpenNIC official websites and search engines, the top three most queried OpenNIC

Rank	Category	IPs	URLs	A Queries
1	Comp. & Internet Info	3,328	15,502	12,992,725
2	Search Engines	2,444	623	2,731,469
3	CDN	2,014	786	2,664,784
4	Business & Economy	1,816	7,069	1,814,833
5	Communications & Telephony	1,350	519	999,506
6	Streaming Media	1,342	765	2,290,172
7	Social Networking	1,336	225	3,230,423
8	Reference & Research	1,307	1,072	1,344,427
9	Shopping	1,266	3,185	1,650,710
10	Storage & Backup	1,060	543	308,898
11	Internet Portals	1,053	81	386,595
12	Shareware & Freeware	1,052	946	243,703
13	Government	1,039	929	68,220
14	News	1,038	2,071	641,388
15	Personal Sites & Blogs	1,035	2,460	1,843,333
30	Adult	496	1,242	114,189
43	Crypto	269	108	33,615
46	Piracy	206	110	6,714
47	Grayware	194	369	6,122
48	Gambling	160	156	1,590
62	Extremism	25	6	3,459
63	Drug Abuse	20	12	43

TABLE V: Total IPs, domains, and A queries for categories.

domains are two personal blogs (by 15 and 11 IPs) and one alternative DNS website (by 11 IPs). We aim to shed light on the purposes of using OpenNIC by benign users. Since most queries are for ICANN domains, we categorize all queried ICANN domains using a domain categorization product by Palo Alto Networks. Table V presents the top categories. Below we further present some interesting observations.

Observation 1: *Accessing sensitive content is not dominating in benign traffic.* Due to the decentralized structure, OpenNIC allows users to bypass ISP censorship or DNS Manipulation, and thus has natural advantages in privacy. A straightforward guess is that one of the main reasons for benign users to use OpenNIC is to access sensitive content, such as porns or gambling websites. However, this guess surprisingly contradicts our data. As Table V shows, only 114,189 A queries from 496 (9.69%) IPs are related to 1,242 domains with the Adult category. Fewer than 206 IPs query domains related to grayware, piracy, drug abuse, or gambling. None of these sensitive contents appear in the top 15 of our data.

Observation 2: *OpenNIC DNS resolvers are utilized in certain countries/regions to bypass Internet blockades.* One example is TikTok, which was reported to be blocked in Venezuela since Jan/6/2025 [88]. TikTok users in Venezuela have to purchase commercial VPNs to bypass the blockage and access TikTok app or services. The first query related to TikTok from users in Venezuela appeared on Jan/26/2025. In total, we observe 202 of 406 source IPs from Venezuela querying domains of TikTok service (e.g., *.tiktokcdn.com). Out of 2,221,911 queries for TikTok domains from Venezuela, 1,284,775 (57.82%) use port 853 via DNS over TLS (DoT), whereas only about 20% of queries for other domains (e.g., *.playstation.com) use DoT. Interestingly, 183 source IPs query domains wikipedia.org and amazon.com at the exact same time (millisecond level) every 20 seconds, which are likely liveness checks by certain VPN software to verify DNS service availability [89], [90].

Observation 3: *Some OpenNIC users are privacy-conscious.*

198,674 A queries from 200 (3.9%) IPs are related to the privacy-enhanced search engine brave.com. Also, 70,628 A queries from 409 (7.99%) IPs are to a private search engine DuckDuckGo. The typical market shares of both are less than 0.8% [91], [92]. In addition, 4,029 A queries from 100 IPs (1.95%) are related to the Tor Project, and 3,760 A queries from 109 IPs (2.13%) are related to I2P networks. Both likely pertain to two common types of the darknet. From AbuseIPDB, 23 source IPs are identified as Tor exit nodes. These Tor nodes may hide DNS queries from public or ISP resolvers for privacy protection.

Our OpenNIC DNS resolver is also utilized by another private DNS server. Since Mar/11/2025, it has recorded 198,105 A and 78,179 AAAA queries, with 23,918 requests seeking A records for domains *.disa.mil and *.dtic.mil, which are managed by DoD Network Information Center. The user identities are concealed by the upper-level DNS resolver.

VII. DISCUSSION

Our study shows that many legitimate users continue to rely on alternative DNS services. Many users who actively choose OpenNIC resolvers do not engage in harmful activities or access sensitive content; rather, they use the Internet normally with more privacy awareness. Many OpenNIC domains serve legitimate purposes, including business and education. Additionally, evidence indicates that OpenNIC resolvers are leveraged by Tor exit nodes, VPNs, and other private DNS services, demonstrating their contribution to a more privacy-enhanced Internet. Meanwhile, botnet activities often target both ICANN and OpenNIC domains. This suggests that simply blocking OpenNIC services would be ineffective and potentially non-compliant. We further discuss several points below.

Domain registration. There is no restriction for OpenNIC domain registration, so everyone can freely register domains and update the records. This can be exploited by botnet creators as a low-cost domain registration alternative. Additionally, anyone can add as many A records as they want for their OpenNIC DNS domains, which similarly can be leveraged by botnet operators. For the TLD .dyn, the API provided by OpenNIC allows users to update the A records of their domains without having to log in to the registration website. This feature provides a simple way for botnet developers to quickly update domain records in the event that the IP addresses of botnet domains are taken down. Based on our communication with OpenNIC administrators, at least one botnet initiator has adopted it in practice.

Defense mechanism. OpenNIC has provided APIs for dynamic blocklists and access control lists. They have also provided best practices for server configuration to block DNS amplification attacks and limit query rates [93]. However, this is not a mandatory requirement for OpenNIC servers. Currently, only 14 of 46 (30.4%) online public Tier-2 servers have setup blocklist mechanisms [35].

DNS amplification. Although this is not specific to OpenNIC, our study also demonstrates that DNS amplification attacks still exist by abusing DNSSEC or TXT records. Besides

forbidding ANY queries, limiting the amount of TXT records is also necessary during domain registration.

VIII. RELATED WORK

DDoS against DNS. DNS resolvers and infrastructure are frequent targets of DDoS attacks [94], [95], [96], [97], such as amplification [98], [99], [100] and random prefix attacks [101]. Rossow showed that 14 UDP protocols are susceptible to bandwidth amplification attacks [29]. Sommese et al. [102] developed methods to map DDoS attacks targeting DNS using 17-month datasets of DNS activities. Li et al. [96] introduced DNSBomb, using DNS features like query aggregation and timeouts to turn low-rate queries into high-volume bursts with amplification factors over 20,000. Xu et al. [97] coordinated vulnerable DNS resolvers to create chained amplification paths, achieving exponential amplification with packet factors exceeding 3,700. Bushart et al. [101] indicated that random prefix attacks may propose chaining-based DoS attacks utilizing CNAME pointers for amplification. Our work has observed several groups of amplification attacks against common websites and one gTLD, as well as random prefix attacks leveraging domains of uncommon websites.

Non-ICANN TLDs. Several prior works have investigated domain misuses with non-ICANN TLDs [103], [104]. For example, the blockchain-based naming systems, such as Namecoin and Emercoin, can easily be abused by C2 servers [104]. It shows that several blockchain domain names have been exploited by malware operators, including .bit domains, which are no longer supported by OpenNIC. Nevertheless, we continue to observe queries for .bit domains on our OpenNIC server. Xia et al. [103] revealed security issues in Ethereum Name Service (ENS), including domain squatting. We observe that domain squatting also exists in OpenNIC and is used for malware spreading. Additionally, Leopard [105] has run an OpenNIC server for a week, and revealed that blockchain domain names can be exploited for C2 servers through techniques such as domain flux. However, that work focuses solely on blockchain-based domains, whereas our study offers a comprehensive analysis covering all DNS queries and botnet-related domains.

DNS Fast Flux, DGA, and Botnets. DNS Fast Flux is a technique where malicious domains quickly change their associated IP addresses, making it difficult for defenders to block. Holz et al. [40] summarized several features of fast flux domains, such as multiple A records in a single lookup. Other works [106], [107] detect fast flux bots using DNS-response features. We also design similar rules to identify suspicious OpenNIC domains from zone files, and further utilize novel DNS query related attributes to detect botnet domains. Several works [26], [23] analyze domain generation algorithms (DGAs) and their adoption on botnets. For example, Plomann et al. [26] conducted a comprehensive measurement study on the DGA landscape, covering 43 DGA-based malware families and variants, which have been integrated in the DGArchive [27] platform to provide domain DGA family lookup and regex guesses.

Antonakakis et al. proposed a line of canonical DNS-telemetry systems for detecting malicious domains, including domain-reputation scoring [108], upper-hierarchy observations [109], and DGA-based botnet detection [110]. Specifically, the Notos [108], a dynamic DNS reputation system, scores previously unseen domains using passive DNS telemetry. It emphasizes network and zone-level characteristics that distinguish malicious agile DNS usage from professional DNS services. In comparison, our goal is not to build a general-purpose reputation framework, but to focus on measurement, detection, and mitigation using lightweight resolver-side query logs. Kopis [109] monitors upper DNS hierarchy at authoritative nameservers and TLD vantage points, and leverages globally visible queries to detect malware domains within a DNS operator’s authority. In contrast, our approach operates at the Tier-2 layer and does not have access to authoritative or TLD nameservers. Pleiades [110] targets DGA-based botnets by monitoring traffic at the local recursive DNS server and analyzing NXDOMAIN responses. It clusters NXDOMAINs based on string similarity and overlapping querying hosts, then applies supervised models to identify infected hosts and pinpoint active DGA-generated C2 candidates. In comparison, our focus is on general botnet-related abuse that goes beyond DGAs, relying on query behaviors and features for identification. Machine learning methods have also been widely used for classifying botnet DNS queries [111], [82], [112], [105]. have investigated machine learning methods for classifying botnet DNS queries. These studies focus on a variety of attributes, including features of domain names, character distribution, DNS requests, DNS responses, source IP addresses, and time sequences. In contrast, we only utilize four features and can achieve similar or even better performance.

IX. CONCLUSION

This paper presents a measurement study on OpenNIC. By operating an OpenNIC DNS server for 208 days and analyzing its domains and query logs, we have identified multiple malicious domains associated with botnet activities. We have also observed that nearly half of the OpenNIC traffic is potentially associated with malicious activity. We have characterized these botnets and developed machine learning models that can accurately identify botnet queries. We have also analyzed benign user activities and presented several interesting observations. Finally, we have reported our findings to OpenNIC administrators for the removal of malicious domains. The anonymized dataset can be requested at <https://github.com/UD-X-Lab/TooOpen>

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